PATHOLOGY

Rapid Revision Notes · NEET PG & INI-CET

How to read these notes :

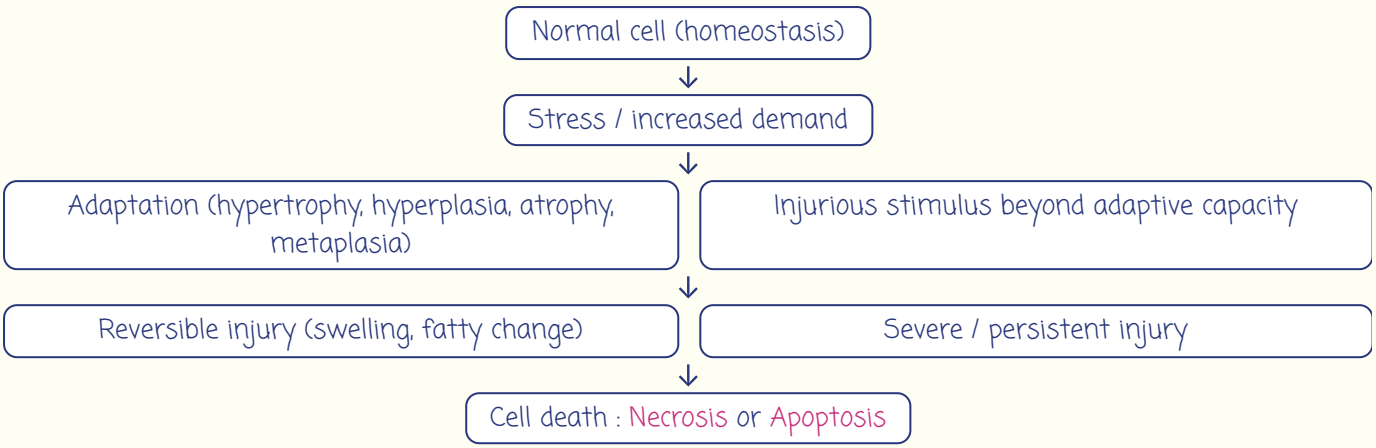
- Navy ink → core content.
- Pink ink → key terms, buzzwords, most-asked facts.
- Dark green ink / green boxes ("Filled") → filled-in high-yield points for a top rank.
- Dashed pink box → mnemonic.
- Yellow box → exam pearl / recent-pattern question.
- Numbered lists → causes, risk factors and criteria, one per line.

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CELL INJURY, ADAPTATION & CELL DEATH

Overview



Cellular Adaptations

Adaptation	Definition	Physiological example	Pathological example
Hypertrophy	↑ cell size → ↑ organ size	Uterus in pregnancy (also hyperplasia), skeletal muscle in body-builders	LVH in hypertension, aortic stenosis
Hyperplasia	↑ cell number (only in dividing cells)	Breast at puberty / lactation, liver regeneration after partial hepatectomy	Endometrial hyperplasia (unopposed estrogen), BPH, warts (HPV)
Atrophy	↓ cell size and number	Thymus after puberty, uterus after delivery	Disuse, denervation, ↓ blood supply, pressure, ↓ nutrition, ↓ endocrine
Metaplasia	One adult differentiated cell type replaced by another	—	Barrett esophagus, squamous metaplasia in smokers' bronchi
Hypoplasia	Incomplete development of an organ (fewer cells)	—	Unilateral renal hypoplasia
Agenesis / Aplasia	Complete absence of organ / its primordium; aplasia = primordium present but no development	—	Renal agenesis (Potter sequence)
Dysplasia	Disordered growth: loss of uniformity and architectural orientation	—	CIN, Barrett with dysplasia

Hypertrophy :

- Seen in permanent cells (cardiac, skeletal muscle) which cannot divide.
- Mechanism : mechanical sensors + growth factors (TGF-β, IGF-1, FGF) + vasoactive agents (α-adrenergic agonists, endothelin-1, angiotensin II).
- Signal pathways : PI3K/AKT (physiological), G-protein coupled receptor signalling (pathological).
- Switch of contractile proteins to fetal forms : α-myosin heavy chain → β-MHC (slower, energy-efficient contraction).
- ↑ ANP (atrial natriuretic peptide) gene expression in hypertrophied ventricle.

Filled

- Pure hypertrophy example : skeletal muscle, heart. Pure hyperplasia : liver regeneration. Both together : gravid uterus (smooth muscle).
- Selective hypertrophy of an organelle : ↑ smooth ER in hepatocytes with barbiturates / phenobarbital (CYP450 induction).
- Cardiac myocyte hypertrophy limit → myofibril fragmentation, loss of contractile elements → cardiac failure.

Hyperplasia :

- Occurs only in tissues capable of division (labile and stable cells).
- Driven by growth factors and stem-cell recruitment.
- Pathological hyperplasia is a fertile soil for cancer : endometrial hyperplasia → endometrial carcinoma; HPV papillomas.
- BPH is NOT pre-malignant for prostatic carcinoma.

Atrophy :

Causes :

1. Decreased workload (disuse atrophy)
2. Loss of innervation (denervation atrophy)
3. Diminished blood supply (senile atrophy of brain)
4. Inadequate nutrition (marasmus, cachexia — TNF-α)
5. Loss of endocrine stimulation (endometrium after menopause)
6. Pressure (erosion of vertebra by aortic aneurysm)

Mechanism :

- ↓ protein synthesis + ↑ protein degradation via ubiquitin-proteasome pathway.
- Autophagy (self-eating) → autophagic vacuoles.
- Residual bodies contain lipofuscin → brown atrophy (heart, liver of elderly).

Metaplasia :

- Reversible; reprogramming of stem cells, not change of differentiated cells.
- Driven by cytokines/growth factors + retinoic acid; transcription factors (SOX, CDX2).

Type	Example	Cause
Columnar → squamous (most common)	Bronchus	Cigarette smoking, vitamin A deficiency
Columnar → squamous	Endocervix (transformation zone)	Low vaginal pH
Transitional → squamous	Urinary bladder	Schistosoma haematobium, stones
Squamous → columnar (intestinal)	Barrett esophagus	GERD (acid reflux)
Gastric → intestinal	Stomach	H. pylori, autoimmune gastritis
Connective tissue	Myositis ossificans (bone in muscle)	Trauma / hemorrhage

Exam pearl

- Epithelial metaplasia is reversible; dysplasia may be reversible (low grade); carcinoma in situ is not invasive but irreversible.
- Myositis ossificans : "zonal pattern" — mature bone at periphery, immature fibroblasts at centre (opposite of osteosarcoma).
- Apocrine metaplasia in fibrocystic breast — not pre-malignant.

Cell Injury — Causes

1. Oxygen deprivation (hypoxia, ischemia) — the most common cause.
2. Physical agents (trauma, temperature, radiation, electric shock).
3. Chemical agents and drugs.
4. Infectious agents.
5. Immunologic reactions.
6. Genetic derangements.
7. Nutritional imbalances.

Filled : Hypoxia vs ischemia

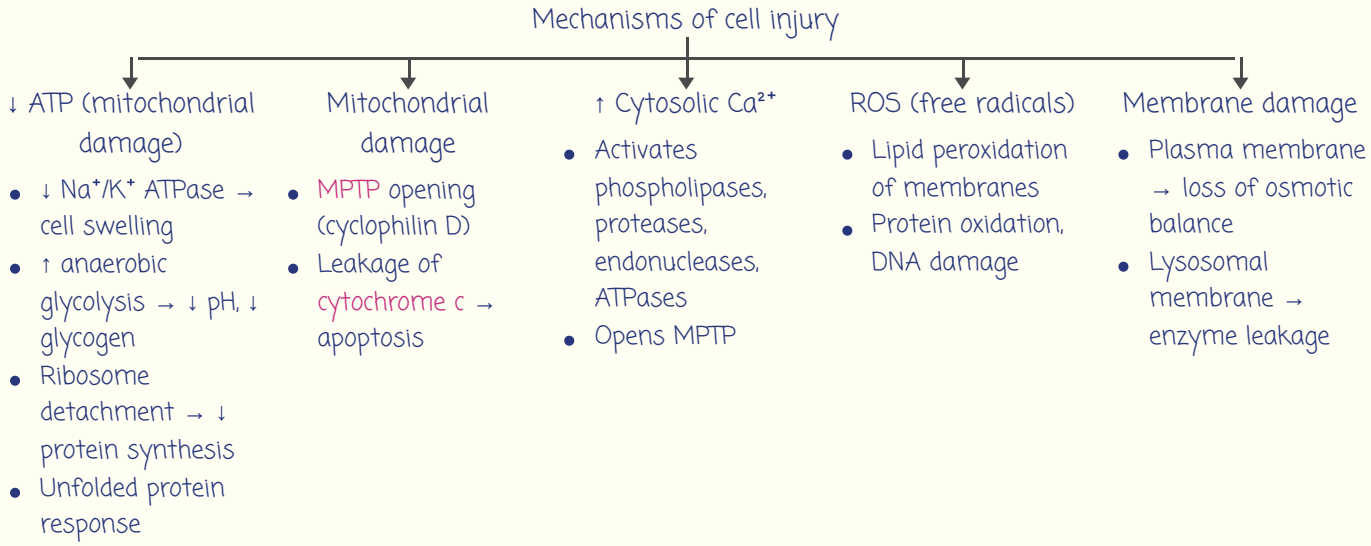
- Ischemia injures tissues faster than hypoxia — loss of both O2 and metabolic substrates (glucose) + accumulation of toxic metabolites.
- In hypoxia, anaerobic glycolysis continues; in ischemia it stops once substrates are exhausted.

Susceptibility of cells to ischemia :

Cell	Time to irreversible injury
Neurons	3-5 min
Myocardium	20-30 min (irreversible); hepatocytes 1-2 h
Fibroblasts, skeletal muscle	Many hours

- Most vulnerable neurons : CA1 of hippocampus (Sommer sector) → Purkinje cells of cerebellum → pyramidal cells of neocortex (layers 3, 5, 6).
- Most sensitive renal segment : proximal tubule (S3 segment) and thick ascending limb.
- Liver : zone 3 (centrilobular) most vulnerable to ischemia.

Mechanisms of Cell Injury



Free radicals (ROS) :

Generation :

1. Normal metabolism (reduction-oxidation reactions)
2. Absorption of radiant energy (UV, X-rays → hydrolysis of water)
3. Leukocyte respiratory burst (NADPH oxidase)
4. Enzymatic metabolism of chemicals (CCl_4 → CCl_3•)
5. Transition metals : Fenton reaction ($Fe^{2+} + H_2O_2 \rightarrow Fe^{3+} + OH• + OH^-$)
6. Nitric oxide (NO) → peroxynitrite ($ONOO^-$)

Removal (antioxidants) :

1. Antioxidants : vitamin E (lipid-soluble, membrane), vitamin A, ascorbic acid, glutathione.
2. Storage/transport proteins that bind free iron and copper : transferrin, ferritin, lactoferrin, ceruloplasmin.
3. Enzymes :
   - Superoxide dismutase : $O_2^- \rightarrow H_2O_2$
   - Catalase (peroxisomes) : $2H_2O_2 \rightarrow O_2 + 2H_2O$
   - Glutathione peroxidase : $H_2O_2 + 2GSH \rightarrow GSSG + 2H_2O$

> Exam pearl
> - Most reactive (damaging) free radical : hydroxyl radical (OH•).
> - Superoxide dismutase is mitochondrial (Mn-SOD) and cytosolic (Cu/Zn-SOD).
> - Free-radical injury examples : CCl_4 hepatotoxicity, reperfusion injury, radiation, oxygen toxicity (retinopathy of prematurity, BPD), acetaminophen (NAPQI), aging.

Ischemia-reperfusion injury :

1. ↑ ROS generation by damaged mitochondria and leukocytes.
2. Intracellular calcium overload (MPTP opening).
3. Inflammation : neutrophil influx with cytokines.
4. Complement activation (IgM deposits in ischemic tissue).

- Morphology in heart : contraction band necrosis (hyper-contracted myofibrils due to Ca^{2+} influx).

Chemical injury :

Direct acting	Indirect (converted to toxic metabolites by CYP450)
Mercuric chloride — binds SH groups (kidney, GI)	CCl_4 → CCl_3• — fatty liver, centrilobular necrosis
Cyanide — inhibits cytochrome oxidase	Acetaminophen → NAPQI — zone 3 necrosis, ↓ glutathione
Chemotherapeutic agents	Bromobenzene, vinyl chloride

Reversible Cell Injury

Two main features :

1. Cellular swelling (hydropic change / vacuolar degeneration) — earliest change; failure of Na^+/K^+ ATPase.
2. Fatty change (steatosis) — liver, heart, kidney (organs in lipid metabolism).

Ultrastructural changes (reversible) :

1. Plasma membrane alterations : blebbing, blunting, loss of microvilli.
2. Mitochondrial changes : swelling, small amorphous densities.
3. Dilatation of ER with detachment of polysomes.
4. Nuclear alterations : disaggregation of granular and fibrillar elements.

5. Myelin figures : phospholipid whorls from damaged membranes.

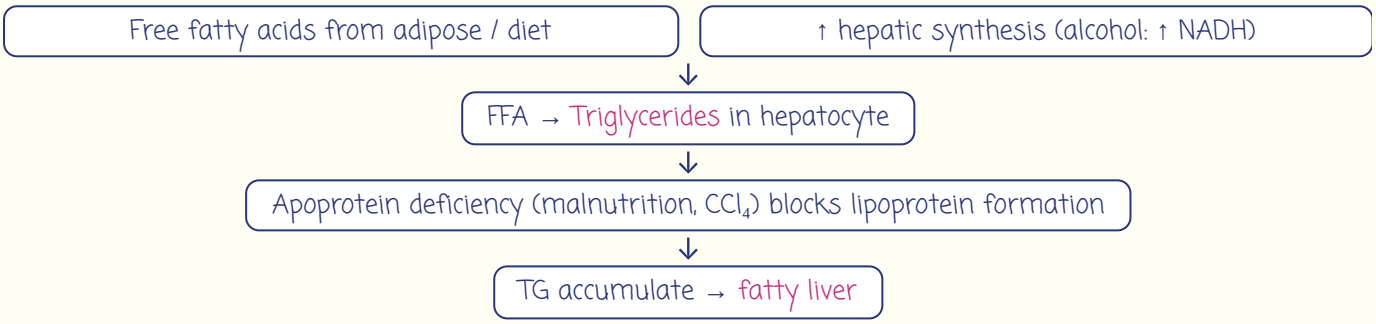
> Filled : Reversible vs irreversible — point of no return
> - Two phenomena characterize irreversibility : (1) inability to restore mitochondrial function (lack of oxidative phosphorylation) even after reperfusion, (2) profound disturbances in membrane function.
> - Large flocculent amorphous densities in mitochondria → irreversible (ischemic myocardium).
> - Leakage of enzymes (troponin, CK-MB, ALT) indicates membrane damage → irreversible injury.

Fatty change (steatosis) — liver :

Causes :

1. Alcohol (most common in developed countries)
2. Non-alcoholic fatty liver disease (obesity, diabetes, metabolic syndrome)
3. Protein malnutrition (kwashiorkor — ↓ apoprotein)
4. Toxins (CCl_4), drugs (tetracycline, valproate)
5. Starvation, hypoxia, pregnancy (acute fatty liver of pregnancy)
6. Reye syndrome (aspirin in children, microvesicular)

Pathogenesis of fatty liver



Feature	Microvesicular	Macrovesicular
Appearance	Many small droplets, central nucleus	Single large vacuole, nucleus pushed to periphery
Causes	Reye syndrome, acute fatty liver of pregnancy, valproate, tetracycline	Alcohol, obesity, diabetes, malnutrition
Significance	Mitochondrial dysfunction; can be severe	Usually benign, reversible

- Stains : fat dissolved in routine paraffin sections (clear vacuoles); demonstrated by Oil Red O and Sudan III/IV on frozen sections; osmium tetroxide (black).
- Heart : fatty change in prolonged moderate hypoxia (profound anemia) → tigered effect / "thrush-breast" heart (bands of yellow myocardium alternating with red-brown) in the papillary muscles.

Hyaline change :

- Homogeneous, glassy, pink (eosinophilic) appearance on H&E — a descriptive term, not a specific lesion.

Intracellular hyaline	Extracellular hyaline
Mallory-Denk bodies (alcoholic hepatitis — keratin)	Hyaline arteriolosclerosis (benign HTN, diabetes)
Russell bodies (immunoglobulin in plasma cells)	Hyaline in old scars, corpora albicantia
Councilman bodies (apoptotic hepatocytes)	Hyaline membranes (ARDS, HMD of newborn)
Hyaline droplets in PCT (proteinuria)	Amyloid (hyaline-like)
Zenker's hyaline degeneration of muscle (typhoid)	Hyaline cartilage-like scar

Other intracellular accumulations :

Substance	Examples
Cholesterol	Atherosclerosis (foam cells), xanthomas, cholesterolosis (strawberry gallbladder), Niemann-Pick type C
Proteins	Russell bodies, Mallory hyaline, α1-antitrypsin globules (PAS +, diastase-resistant)
Glycogen	Diabetes (Armanni-Ebstein lesion — PCT/loop of Henle), glycogen storage diseases (stain : PAS + diastase-sensitive, Best carmine)
Pigments	Lipofuscin, melanin, hemosiderin, bilirubin, carbon (see Pigments chapter)

Irreversible Injury — Necrosis

- Necrosis : death of cells in a living organism due to severe injury, always pathological, with inflammation.

Morphology of necrotic cell :

- Cytoplasm : ↑ eosinophilia (loss of RNA + denatured proteins), glassy, myelin figures, calcification.
- Nucleus :

Pyknosis (shrinkage, ↑ basophilia) → Karyorrhexis (fragmentation) → Karyolysis (fading; DNase)

Patterns of necrosis :

Type	Mechanism / features	Classic examples
Coagulative	Protein denaturation > enzymatic digestion; architecture preserved for days ("ghost cells")	Infarcts of all solid organs except brain (heart, kidney, spleen)
Liquefactive	Enzymatic digestion dominates; tissue becomes liquid, creamy (pus)	Brain infarct, abscess (bacterial/fungal)
Caseous	"Cheese-like", friable white; granuloma with central amorphous granular debris	Tuberculosis, histoplasmosis, coccidioidomycosis
Fat necrosis	Lipases release FA → combine with Ca^{2+} (saponification) → chalky white	Acute pancreatitis, traumatic fat necrosis of breast
Fibrinoid	Immune complexes + fibrin in vessel walls; bright pink, amorphous	Malignant hypertension, vasculitis (PAN), Aschoff body, preeclampsia
Gangrenous (clinical term)	Usually coagulative necrosis of limb (dry), + bacterial liquefaction (wet)	Diabetic foot, intestinal gangrene

Filled : Extra high-yield

- Gumma (tertiary syphilis) — rubbery necrosis, architecture of vessels partially preserved (ghost outlines) — vs caseation where outlines are lost.
- Caseation without TB : histoplasmosis, coccidioidomycosis, sarcoid (rarely, no), tularemia, cat-scratch (stellate microabscess).
- Fibrinoid necrosis in placenta (normal finding at term — Nitabuch layer).
- Fat necrosis in breast can mimic carcinoma clinically (hard lump + skin retraction) — history of trauma.

Apoptosis

- Programmed cell death; membrane intact, no inflammation.
- Hallmark morphology : cell shrinkage, chromatin condensation (pyknosis, most characteristic), cytoplasmic blebs, apoptotic bodies, phagocytosis by macrophages.
- DNA fragmentation : step-ladder pattern on agarose gel (internucleosomal, 180-200 bp multiples) vs smear pattern in necrosis.

Causes :

Physiological :

1. Embryogenesis (implantation, organogenesis, involution — e.g., interdigital webs)
2. Hormone-dependent involution (endometrium in menstruation, ovarian follicular atresia, breast after weaning)
3. Cell turnover in proliferating populations (intestinal crypt epithelium)
4. Elimination of self-reactive lymphocytes (negative selection in thymus)
5. Death of host cells that have served their purpose (neutrophils after acute inflammation)

Pathological :

1. DNA damage (radiation, cytotoxic drugs)
2. Accumulation of misfolded proteins (ER stress)
3. Infections — viral hepatitis (Councilman bodies)
4. Pathological atrophy in parenchymal organs after duct obstruction (pancreas, kidney, parotid)
5. Cell death by cytotoxic T cells (graft rejection)

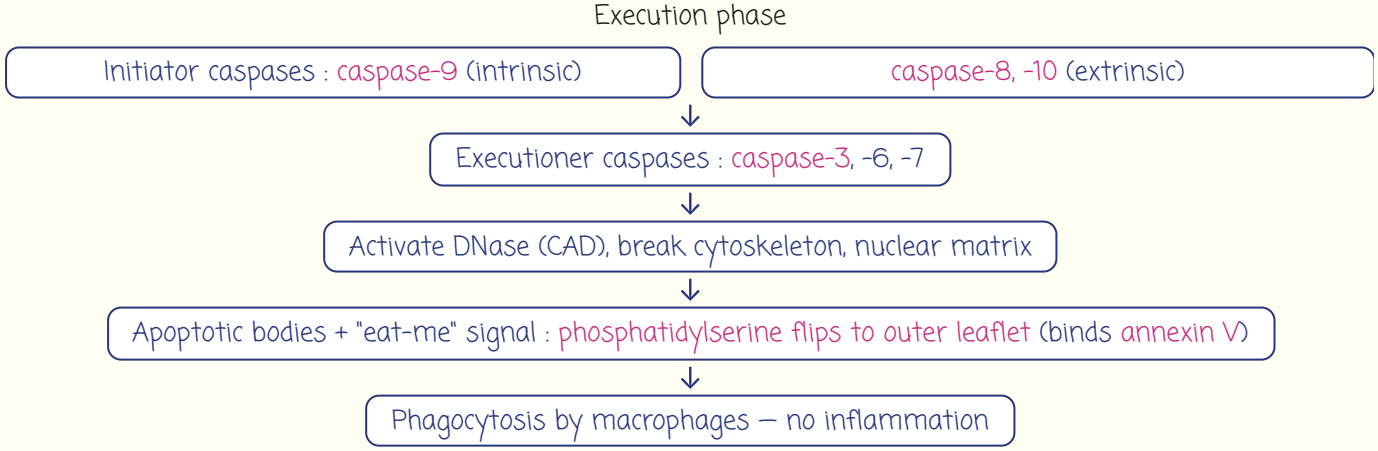
Pathways :

Intrinsic (mitochondrial) pathway :

- Most common pathway.
- Trigger : ↓ growth factors, DNA damage (p53), misfolded proteins.
- BH3-only sensors (BIM, BID, BAD, PUMA, NOXA) ↑ → activate BAX/BAK (pro-apoptotic).
- ↓ BCL-2, BCL-XL, MCL-1 (anti-apoptotic).
- Cytochrome c release → binds APAF-1 → apoptosome → caspase-9 (initiator).

Extrinsic (death receptor) pathway :

- Death receptors : TNFR1, Fas (CD95) — have death domain.
- FasL (on activated T cells) binds Fas → adaptor FADD → caspase-8 (also caspase-10).
- Inhibitor : FLIP (viral and cellular).
- Role : eliminating self-reactive lymphocytes, CTL killing.
- Defect → autoimmune lymphoproliferative syndrome (ALPS).



Filled : Rank boosters — apoptosis

- Caspase = cysteine protease that cleaves after aspartic acid.
- BCL-2 overexpression by t(14;18) → follicular lymphoma (failure of apoptosis).
- IAP (inhibitor of apoptosis proteins) block caspases; SMAC/DIABLO from mitochondria neutralize IAP.
- Markers of apoptosis : annexin V, TUNEL assay, caspase-3 cleavage, DNA ladder.
- Granzyme B from CTL activates caspases directly (and BID); perforin makes pores.
- p53 → PUMA/NOXA → BAX/BAK; also arrests cell cycle via p21.

Necrosis vs Apoptosis :

Feature	Necrosis	Apoptosis
Cell size	Enlarged (swelling)	Reduced (shrinkage)
Nucleus	Pyknosis → karyorrhexis → karyolysis	Fragmentation into nucleosome-size pieces
Plasma membrane	Disrupted	Intact; altered structure (lipid orientation)
Cellular contents	Enzymatic digestion; may leak out	Intact; released in apoptotic bodies
Inflammation	Frequent	None
Role	Always pathological	Often physiological; may be pathological
DNA on gel	Smear	Ladder
ATP	Not required	Energy-dependent

Other Forms of Cell Death

Type	Key molecules	Features / examples
Necroptosis	RIPK1, RIPK3 → MLKL	Programmed necrosis, caspase-independent; TNF; Crohn disease, pancreatitis, reperfusion; CMV inhibits
Pyroptosis	Inflammasome → caspase-1 (also 4, 5, 11) → gasdermin D pores	Fever (IL-1β, IL-18); infected cells (Salmonella, Shigella)
Ferroptosis	Iron-dependent lipid peroxidation; ↓ GPX4 / glutathione	Stroke, neurodegeneration, cancer; inhibited by ferrostatin
Autophagy	Beclin-1, LC3, ATG genes; mTOR inhibition	Starvation survival; defects in Crohn (ATG16L1), Alzheimer, Huntington
Entosis	Cell engulfed by neighbour	"Cell-in-cell" in tumours
Anoikis	Loss of integrin-matrix attachment	Prevents metastasis; resistance in cancer
NETosis	Neutrophil extracellular traps (PAD4)	Sepsis, SLE, thrombosis
Parthanatos	PARP-1 overactivation, AIF	Stroke, DNA damage

Mnemonic : Ferroptosis

"Ferro = Fe, peroxidation, GPX4 guards" — iron + lipid ROS; not apoptosis (no caspases).

Autophagy

- "Self-eating" — delivers cytoplasmic materials to lysosomes.
- Types : macroautophagy (main), microautophagy, chaperone-mediated autophagy.
- Steps : nucleation (Beclin-1 / VPS34) → elongation (ATG genes, LC3) → autophagosome → fusion with lysosome → autophagolysosome.
- Induced by nutrient deprivation (mTOR inhibition).

Cellular Aging

1. DNA damage (accumulated mutations; defective repair — Werner syndrome: DNA helicase, Bloom syndrome).
2. Replicative senescence — telomere shortening (Hayflick limit); telomerase active in germ/stem/cancer cells.
3. Defective protein homeostasis (↓ chaperones, ↓ proteasome).
4. Dysregulated nutrient sensing : caloric restriction ↑ lifespan via ↓ IGF-1 and ↓ mTOR, ↑ sirtuins (NAD-dependent deacetylases).
5. Persistent inflammation ("inflammaging"), ROS.

Filled : Progeria syndromes

Syndrome	Defect
Werner syndrome	WRN — DNA helicase (RecQ) — adult-onset premature aging
Hutchinson-Gilford progeria	LMNA (lamin A) — progerin — childhood aging
Bloom syndrome	BLM helicase
Ataxia telangiectasia	ATM kinase — radiation sensitivity
Dyskeratosis congenita	Telomerase (TERC/TERT/DKC1)

CALCIFICATION & AMYLOIDOSIS

Pathological Calcification

- Abnormal tissue deposition of calcium salts with small amounts of iron, magnesium and other minerals.
- Stain : von Kossa (black — actually stains phosphate/carbonate), Alizarin red S (red-orange, calcium). On H&E : basophilic, amorphous, granular.

Feature	Dystrophic	Metastatic
Serum calcium	Normal	Hypercalcemia
Tissue	Dead / dying (necrotic) tissue	Normal (viable) tissue
Common sites	Atheroma, damaged / aging heart valves, caseous TB, infarcts, dead parasites	Interstitial tissue of gastric mucosa, kidneys, lungs, systemic arteries, pulmonary veins
Mechanism	Initiation in membrane-bound vesicles (phospholipids + phosphatase) or extracellular	Loss of acid (H^+) → alkaline compartment favours deposition
Clinical	Usually a marker of prior injury	Nephrocalcinosis, respiratory deficits

Mnemonic : Sites of metastatic calcification

"Calcium loves alkaline tissues" — tissues that excrete acid have an internal alkaline compartment : gastric mucosa (HCl), kidney (H^+), lung (CO_2).

Causes of metastatic calcification (hypercalcemia) :

1. ↑ Parathyroid hormone — primary hyperparathyroidism, PTHrP from tumours (squamous cell lung carcinoma)
2. Bone destruction — multiple myeloma, leukemia, skeletal metastases, Paget disease (accelerated turnover), immobilization
3. Vitamin D-related — vitamin D intoxication, sarcoidosis (macrophages activate vitamin D), idiopathic hypercalcemia of infancy (Williams syndrome)
4. Renal failure — phosphate retention → secondary hyperparathyroidism
5. Milk-alkali syndrome (excess calcium + absorbable antacids)
6. Aluminium intoxication (chronic dialysis)

Dystrophic calcification — examples :

1. Atherosclerotic plaques (advanced atheroma)
2. Aging or damaged heart valves (calcific aortic stenosis, bicuspid aortic valve)
3. Caseous necrosis in TB (Ghon complex, calcified lymph nodes)
4. Fat necrosis (pancreatitis — "chalky white")
5. Mönckeberg medial calcific sclerosis (medium muscular arteries, > 50 y)
6. Dead parasites — hydatid cyst wall, cysticercus, Schistosoma eggs
7. Thrombi (phleboliths in veins)
8. Tumours — psammoma bodies

Psammoma bodies :

- Laminated, concentric calcified structures (dystrophic) formed by progressive deposition on single necrotic cells.

Mnemonic : Psammoma bodies — "PSaMMOMa"

Papillary carcinoma of thyroid · Serous cystadenocarcinoma of ovary · Meningioma · Malignant Mesothelioma · also prolactinoma, papillary RCC, somatostatinoma (duodenum), endometrial serous carcinoma.

Filled

- Calciphylaxis (calcific uremic arteriolopathy) — end-stage renal disease, ↑ Ca × P product, medial calcification of small arteries → painful skin necrosis.
- Calcinosis cutis in CREST / limited scleroderma, dermatomyositis (children).
- Heterotopic ossification = bone formation (not just calcium) in soft tissue — myositis ossificans.
- Michaelis-Gutmann bodies (target-like, calcified, PAS+, von Kossa +) — malakoplakia of bladder (defective macrophage lysosomal killing of E. coli).
- Schaumann bodies (laminated calcium + protein) and asteroid bodies — sarcoidosis.

Amyloidosis

- Extracellular deposition of misfolded proteins which aggregate into insoluble fibrils with a β-pleated sheet configuration.
- All amyloid : ~95% fibril protein + 5% P component (serum amyloid P, pentraxin) + glycosaminoglycans (heparan sulfate).

Physical nature :

- EM : non-branching fibrils, 7.5-10 nm diameter, indefinite length.
- X-ray crystallography / infrared spectroscopy : cross-β-pleated sheet conformation (responsible for Congo red birefringence).

Staining :

Stain	Result
H&E	Pale eosinophilic, amorphous, hyaline extracellular material
Congo red (light microscopy)	Pink / red (salmon-pink)
Congo red under polarized light	Apple-green birefringence (most specific)
Gross (Lugol's iodine)	Mahogany brown → blue-violet with dilute H_2SO_4 (starch-like, hence "amyloid" by Virchow)
Thioflavin T / S	Fluorescence (UV)
Crystal violet / methyl violet	Metachromasia (rose-pink on violet)
Immunohistochemistry / mass spectrometry	Typing of amyloid (gold standard for type)

Classification of amyloid proteins :

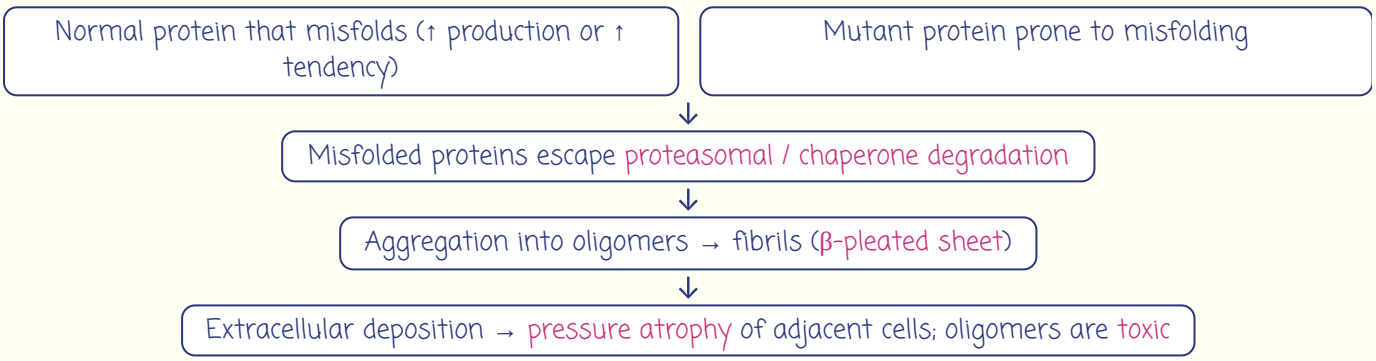
Protein	Precursor	Clinical setting
AL (amyloid light chain)	Ig light chains (λ > κ)	Primary / plasma cell dyscrasia — multiple myeloma, B-cell neoplasms
AA (amyloid-associated)	SAA (acute phase, liver; IL-1/IL-6)	Reactive systemic (secondary) — chronic inflammation
Aβ2m (β2-microglobulin)	β2-microglobulin (MHC-I)	Long-term hemodialysis — carpal tunnel, joints
ATTR	Transthyretin (normal or mutant)	Familial amyloid polyneuropathy; senile systemic amyloidosis (heart, wild-type TTR)
Aβ (amyloid β)	APP (amyloid precursor protein)	Alzheimer disease, cerebral amyloid angiopathy
ACal	(Pro)calcitonin	Medullary carcinoma thyroid

Protein	Precursor	Clinical setting
AIAPP	Amylin (islet amyloid polypeptide)	Type 2 diabetes (islets)
AANF	Atrial natriuretic factor	Isolated atrial amyloidosis (elderly)
APrP	Prion protein	Spongiform encephalopathies
AA (mutant pyrin)	SAA	Familial Mediterranean fever (MEFV — pyrin)
ALys / AFib / AApoA1	Lysozyme, fibrinogen, apo A1	Hereditary renal amyloidosis

Causes of reactive systemic (AA) amyloidosis :

1. Tuberculosis (most common in India), bronchiectasis, chronic osteomyelitis, lepromatous leprosy
2. Rheumatoid arthritis (most common in West), ankylosing spondylitis
3. Inflammatory bowel disease (Crohn > UC)
4. Skin popping (s.c. heroin abuse)
5. Non-immunocyte tumours : renal cell carcinoma, Hodgkin lymphoma
6. Familial Mediterranean fever, other auto-inflammatory syndromes

Pathogenesis :



Exam pearl

- Primary AL amyloid : most common form in developed countries; occurs in 5-15% of multiple myeloma.
- Bence Jones protein (free light chains) in urine/serum in AL amyloidosis — even without overt myeloma ("monoclonal gammopathy").
- AL amyloidosis = Congo red + ; AA = Congo red loses affinity after potassium permanganate ($KMnO_4$) treatment; AL is permanganate-resistant.

Morphology (organ involvement) :

Organ	Features
Kidney (most common, most serious)	Enlarged, pale, waxy; deposits in glomeruli (mesangium, subendothelial) → nephrotic syndrome; later shrunken. Peritubular and vessel walls.
Spleen	Sago spleen (deposits in splenic follicles — tapioca-like granules); Lardaceous spleen (deposits in red pulp / sinusoid walls — map-like).
Liver	Hepatomegaly; deposits in space of Disse → pressure atrophy of hepatocytes; liver function usually preserved.
Heart	Restrictive cardiomyopathy, conduction disturbances, low voltage ECG with thick walls; "sparkling / granular" echo; most common cause of death in AL amyloid. Subendocardial deposits.

Organ	Features
Others	Macroglossia (AL), carpal tunnel syndrome, adrenal insufficiency, GI malabsorption, periorbital purpura ("raccoon eyes", AL), pinch purpura

Diagnosis — biopsy sites :

1. Abdominal fat pad aspiration — preferred screening (simple, safe)
2. Rectal biopsy (submucosal vessels) — 75% positive
3. Gingival / labial salivary gland biopsy
4. Renal biopsy — most sensitive in renal disease
5. Bone marrow biopsy (AL)
6. Serum / urine immunofixation and free light chain assay; scintigraphy (SAP scan, Tc-PYP for ATTR cardiac)

Filled : Rank boosters

- Localized amyloidosis : nodular lung, larynx, skin, bladder, tongue; endocrine amyloid (medullary Ca thyroid, islet in T2DM, pituitary prolactinoma).
- Senile cardiac amyloidosis (wild-type ATTR) — men > 70 y; ATTR V122I mutation in 4% African Americans.
- Drug for ATTR : Tafamidis (TTR stabilizer), Patisiran (siRNA), Inotersen.
- β2-microglobulin is not removed by cuprophane dialysis membranes.
- Amyloid in Alzheimer : Aβ42 — neuritic plaques; APP gene on chromosome 21 → early Alzheimer in Down syndrome.
- Hereditary : Familial Mediterranean fever — autosomal recessive, MEFV (pyrin), AA amyloid; Muckle-Wells (cryopyrin) — AA.
- Most common cause of death in systemic amyloidosis overall : renal failure (AA) / cardiac (AL).

INFLAMMATION

Overview

- Protective response involving host cells, blood vessels, proteins and mediators, intended to eliminate the initial cause and the necrotic cells, and initiate repair.
- Cardinal signs : Rubor, Calor, Tumor, Dolor (Celsus) + Functio laesa (loss of function — Virchow).

Feature	Acute	Chronic
Onset	Fast : minutes-hours	Slow : days
Cellular infiltrate	Mainly neutrophils	Monocytes / macrophages, lymphocytes, plasma cells
Tissue injury, fibrosis	Usually mild, self-limited	Often severe and progressive
Local & systemic signs	Prominent	Less
Hallmark	Exudation (edema)	Tissue destruction + repair (angiogenesis, fibrosis) simultaneously

Stimuli for acute inflammation :

1. Infections (bacterial, viral, fungal, parasitic) and microbial toxins
2. Tissue necrosis (ischemia, trauma, burns)
3. Foreign bodies (splinters, sutures, urate / cholesterol crystals)
4. Immune reactions (hypersensitivity, autoimmunity)

Recognition of microbes and damaged cells :

Receptor	Ligand / function
Toll-like receptors (TLRs)	PAMPs; TLR4 → LPS (with CD14), TLR2 → peptidoglycan/lipoteichoic acid, TLR5 → flagellin, TLR3 → dsRNA, TLR7/8 → ssRNA, TLR9 → CpG DNA
NOD-like receptors / inflammasome (NLRP3)	DAMPs — urate crystals (gout), cholesterol, ATP, K^+ efflux → caspase-1 → IL-1β
C-type lectin receptors	Fungal glycans (Dectin-1)
RIG-like receptors	Viral RNA (cytosol)

Filled

- Gain-of-function NLRP3 (cryopyrin) mutations → cryopyrin-associated periodic syndromes (Muckle-Wells, FCAS) → treated with anakinra (IL-1R antagonist).
- TLRs : all are on cell surface except TLR3, 7, 8, 9 (endosomal — nucleic acids).

Vascular Events

1. Transient vasoconstriction (seconds) — first, neurogenic.
2. Vasodilation (histamine, NO) — erythema, warmth (earliest consistent change).
3. Increased vascular permeability → exudate → hallmark of acute inflammation.
4. Stasis (slowing of circulation, ↑ viscosity) → margination of leukocytes.

Mechanisms of increased vascular permeability :

Mechanism	Mediators	Vessel	Timing
Endothelial cell contraction (most common)	Histamine, bradykinin, leukotrienes, substance P	Post-capillary venules	Immediate transient (15-30 min)

Mechanism	Mediators	Vessel	Timing
Endothelial retraction (cytoskeletal reorganization)	IL-1, TNF, IFN-γ	Venules	Delayed (4–6 h), prolonged (24 h+)
Direct endothelial injury	Burns, toxins	All (venules, capillaries, arterioles)	Immediate sustained
Delayed prolonged leakage	Mild thermal injury, X-rays, UV (sunburn), some bacterial toxins	Venules + capillaries	Begins 2–12 h, lasts hours–days
Leukocyte-mediated injury	ROS, proteases	Venules, pulmonary capillaries	Late
Transcytosis (VVO)	VEGF	Venules	—
Leakage from new vessels	VEGF (angiogenesis)	Regenerating vessels	Until mature

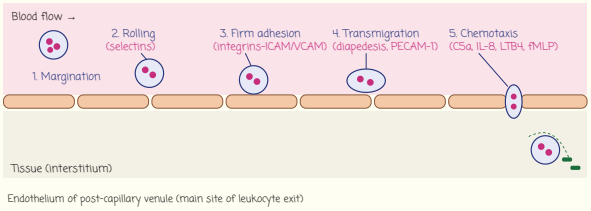
Exudate vs transudate :

Feature	Exudate	Transudate
Mechanism	↑ vascular permeability (inflammation)	↑ hydrostatic / ↓ oncotic pressure
Protein	> 3 g/dL	< 3 g/dL
Specific gravity	> 1.020	< 1.012
Cells	Many (inflammatory)	Few
Light's criteria (pleural)	Fluid/serum protein > 0.5; fluid/serum LDH > 0.6; fluid LDH > 2/3 upper limit of normal serum LDH	Not met

Exam pearl

- Lymphatics proliferate in inflammation; lymphangitis (red streaks) + lymphadenitis — e.g., streptococcal cellulitis.
- Edema due to inflammation = exudate; pus = purulent exudate rich in neutrophils, dead cells, microbes.

Cellular Events — Leukocyte Recruitment



Steps of leukocyte extravasation (labelled)

Adhesion molecules :

Endothelial molecule	Leukocyte ligand	Role
P-selectin (CD62P) — from Weibel-Palade bodies (histamine, thrombin)	Sialyl-Lewis X (PSGL-1)	Rolling
E-selectin (CD62E) — induced by IL-1, TNF	Sialyl-Lewis X	Rolling, adhesion
GlyCam-1, CD34	L-selectin (CD62L)	Rolling (homing of lymphocytes to HEV)
ICAM-1 (Ig family)	LFA-1 (CD11a/CD18), Mac-1 (CD11b/CD18) — β2 integrins	Firm adhesion, arrest, transmigration

Endothelial molecule	Leukocyte ligand	Role
VCAM-1	VLA-4 (β1 integrin)	Adhesion (eosinophils, monocytes, lymphocytes)
PECAM-1 (CD31)	PECAM-1	Transmigration (diapedesis)

- Integrin affinity ↑ by chemokines (IL-8) displayed on endothelium ("inside-out" signalling).
- Leukocytes pierce basement membrane by collagenases.
- Neutrophils dominate first 6-24 h, replaced by monocytes in 24-48 h (exception : Pseudomonas — neutrophils persist; viral — lymphocytes first; allergy — eosinophils).

Leukocyte adhesion deficiency (LAD) :

Type	Defect	Features
LAD-1	β2 integrin (CD18) — ITGB2	Delayed separation of umbilical cord (> 30 days), recurrent bacterial infections without pus, neutrophilia, periodontitis
LAD-2	Sialyl-Lewis X (fucosyl transferase — fucose metabolism)	Milder infections, Bombay blood group, mental retardation, short stature
LAD-3	Kindlin-3 (integrin activation)	LAD-1 like + bleeding (Glanzmann-like)

Chemotaxis :

- Exogenous : N-formyl methionine peptides (bacterial), lipids.
- Endogenous : C5a, LTB4, IL-8 (CXCL8), kallikrein, platelet activating factor.
- Receptors : 7-transmembrane G-protein coupled; movement via actin polymerization at leading edge (filopodia).

Phagocytosis & killing :

Recognition & attachment (opsonins : IgG Fc, C3b, collectins/MBL) → Engulfment (phagosome) → Fusion with lysosome (phagolysosome) → Killing & degradation

Killing mechanism	Details
Oxygen-dependent (respiratory burst)	NADPH oxidase (phagocyte oxidase) : $O_2 \rightarrow O_2^- \rightarrow H_2O_2$; Myeloperoxidase (MPO) : $H_2O_2 + Cl^- \rightarrow HOCl$ (hypochlorous acid / bleach) — most potent bactericidal system
Nitric oxide	iNOS (macrophages) → NO + O_2^- → peroxynitrite
Oxygen-independent	Lysozyme, lactoferrin, major basic protein (eosinophils), defensins, BPI, elastase, cathelicidins
NETs	Extracellular chromatin + granule proteins (PAD4 → citrullinated histones); anti-NET antibodies in SLE

Defects in leukocyte function :

Disease	Defect	Key features
Chronic granulomatous disease (CGD)	NADPH oxidase (gp91phox — X-linked, CYBB, 70%; autosomal recessive p47phox)	Infections with catalase-positive organisms (S. aureus, Aspergillus, Nocardia, Serratia, Burkholderia); granulomas. Test : NBT test (no blue), dihydrorhodamine (DHR) flow cytometry (preferred)
Chédiak-Higashi syndrome	LYST — lysosomal trafficking (AR)	Giant granules in neutrophils, partial albinism, peripheral neuropathy, bleeding, NK defect

Disease	Defect	Key features
Myeloperoxidase deficiency	MPO	Most common inherited phagocyte defect; mostly asymptomatic; Candida infections in diabetics; NBT normal
LAD 1/2	Integrins / selectin ligands	See above
Hyper-IgE (Job) syndrome	STAT3 (AD)	Cold staphylococcal abscesses, eczema, ↑ IgE, retained primary teeth, coarse facies, ↓ Th17

Mnemonic : Catalase-positive organisms in CGD — "Cats Need PLACESS"

Nocardia, Pseudomonas, Listeria, Aspergillus, Candida, E. coli, Staph aureus, Serratia (+ Burkholderia cepacia).

Chemical Mediators

Mediator	Source	Principal actions
Histamine	Mast cells, basophils, platelets	Vasodilation, ↑ permeability (immediate), endothelial activation
Serotonin (5-HT)	Platelets, enterochromaffin cells	Vasoconstriction, ↑ permeability
PGI_2 (prostacyclin)	Endothelium	Vasodilation, inhibits platelet aggregation
TXA_2	Platelets	Vasoconstriction, platelet aggregation
PGE_2	Mast cells, leukocytes	Pain, fever, vasodilation
PGD_2	Mast cells	Vasodilation, ↑ permeability, chemotaxis
LTB_4	Leukocytes	Chemotaxis, leukocyte activation
LTC_4, D_4, E_4 (SRS-A)	Leukocytes, mast cells	Bronchospasm, ↑ permeability, vasoconstriction
Lipoxins (LXA_4, B_4)	Leukocytes + platelets (transcellular)	Anti-inflammatory — inhibit neutrophil recruitment
PAF	Leukocytes, mast cells	Platelet aggregation, bronchospasm, ↑ permeability, vasodilation (low dose)
TNF, IL-1	Macrophages, endothelium, mast cells	Endothelial activation, fever, acute phase response, cachexia (TNF)
IL-6	Macrophages	Acute phase proteins (liver)
Chemokines (IL-8)	Leukocytes, macrophages	Chemotaxis
Bradykinin	Plasma (kininogen)	↑ Permeability, pain, vasodilation, bronchoconstriction
C3a, C5a	Plasma (complement)	Anaphylatoxins → mast cell histamine release; C5a : chemotaxis
C3b	Complement	Opsonin
NO	Endothelium (eNOS), macrophages (iNOS)	Vasodilation, microbial killing, ↓ platelet adhesion

Arachidonic acid pathway :

Membrane phospholipids

↓

↓ Phospholipase A2 (inhibited by corticosteroids via lipocortin / annexin A1)

↓

Arachidonic acid

↓

Cyclooxygenase (COX-1, COX-2) — blocked by NSAIDs, aspirin (irreversible)

5-Lipoxygenase — blocked by zileuton

↓

PGG_2 → PGH_2 → PGI_2, TXA_2, PGD_2, PGE_2, $PGF_2\alpha$

5-HPETE → LTA_4 → LTB_4 ; LTC_4 → D_4 → E_4 (receptor blocked by montelukast) ; lipoxins (12-LO)

Filled

- Aspirin acetylates COX → also triggers aspirin-triggered lipoxins (15-epi-LX) — anti-inflammatory.
- Resolvins, protectins (from omega-3 fatty acids) → resolution of inflammation.
- Aspirin-exacerbated respiratory disease (Samter triad) : asthma + nasal polyps + aspirin sensitivity — shunting to LT pathway.
- Most important mediators of fever : IL-1, TNF, IL-6 → COX in hypothalamus → PGE_2 (reset thermostat).

Plasma protease systems :

Hageman factor (factor XII) activated by collagen, basement membrane, activated platelets

↓

Kinin system : prekallikrein → kallikrein → bradykinin (degraded by ACE / kininase)

Clotting system : thrombin → fibrinopeptides

Fibrinolytic : plasmin → fibrin split products, C3 cleavage

↓

Complement activation (plasmin, kallikrein)

- ACE inhibitors → ↑ bradykinin → cough, angioedema.
- C1-inhibitor deficiency → hereditary angioedema (↑ bradykinin; ↓ C4).

Complement system :

Pathway	Trigger	Key steps
Classical	Antigen-antibody (IgM > IgG3 > IgG1), C1q	C1 → C4, C2 → C3 convertase (C4b2a)
Alternative	Microbial surfaces (LPS), spontaneous C3 hydrolysis	C3b + factor B, factor D → C3bBb (stabilized by properdin)
Lectin	Mannose-binding lectin on microbes	MASP → C4, C2

- Common : C3 convertase → C5 convertase → C5b-C9 (MAC) → lysis.
- Regulators : C1 inhibitor, DAF (CD55) and CD59 (MIRL) (↓ in PNH), factor H, factor I.

Deficiency	Result
C1q, C2, C4 (early classical)	SLE-like immune complex disease (C2 = most common deficiency)
C3	Severe pyogenic infections, immune complex GN
C5-C9 (MAC)	Recurrent Neisseria infections
C1 inhibitor	Hereditary angioedema
DAF / CD59	PNH

Deficiency	Result
Factor H / I	Atypical HUS, C3 glomerulopathy

Morphologic Patterns of Acute Inflammation

Pattern	Features	Examples
Serous	Thin, cell-poor fluid	Skin blister (burn), serous effusions, viral
Fibrinous	Fibrin deposition in body cavities	Bread-and-butter pericarditis (rheumatic, uremic), lobar pneumonia, fibrinous pleuritis
Suppurative / purulent	Pus — neutrophils, necrotic debris	Abscess (S. aureus), acute appendicitis
Ulcer	Local defect / excavation of surface	Peptic ulcer, aphthous ulcer
Pseudomembranous	Necrotic epithelium + fibrin + exudate membrane	C. difficile colitis, diphtheria
Catarrhal	Mucus hypersecretion	Common cold, allergic rhinitis
Hemorrhagic	Exudate with RBCs	Plague, anthrax, meningococcal septicemia, influenza pneumonia
Membranous (croupous)	Fibrin + necrotic cells on surface	Diphtheria (also pseudomembranous)
Cellulitis	Diffuse spread through tissue planes	Streptococcus pyogenes (hyaluronidase, streptokinase)

Exam pearl

- Abscess : central necrosis + neutrophils, surrounded by zone of preserved neutrophils→ vascular dilatation→ fibroblast proliferation (pyogenic membrane).
- Outcome of acute inflammation : resolution, chronic inflammation, fibrosis (scarring), abscess.

Chronic Inflammation

Causes :

1. Persistent infections — TB, Treponema, fungi, viruses, parasites (delayed-type hypersensitivity)
2. Hypersensitivity diseases — autoimmune (RA, IBD), allergic (asthma)
3. Prolonged exposure to toxic agents — exogenous (silica), endogenous (cholesterol in atherosclerosis)
4. Some diseases not conventionally inflammatory — Alzheimer, metabolic syndrome, cancer

Cells of chronic inflammation :

Cell	Key points
Macrophage (dominant cell)	Monocyte → tissue; half-life of monocyte ~1 day, macrophage months-years. M1 (classical) : IFN-γ, LPS → microbicidal (NO, ROS). M2 (alternative) : IL-4, IL-13 → repair, fibrosis (TGF-β), anti-inflammatory (IL-10)
Lymphocytes	Th1 (IFN-γ)→ macrophage activation; Th2 (IL-4, 5, 13) → eosinophils; Th17 (IL-17) → neutrophils
Plasma cells	Antibody production; Russell bodies; "cartwheel / clock-face" chromatin
Eosinophils	Eotaxin (CCL11) chemotaxis; major basic protein — toxic to parasites, epithelium
Mast cells	FcεRI (IgE); histamine — allergy, anaphylaxis
Neutrophils	Chronic osteomyelitis, chronic lung damage (smoking) — "acute on chronic"

Tissue macrophage names :

Tissue	Name
Liver	Kupffer cells
Spleen, lymph nodes	Sinus histiocytes
Lung	Alveolar macrophages (dust cells)
CNS	Microglia
Bone	Osteoclasts
Skin	Langerhans cells (dendritic; Birbeck granules, CD1a, S100, langerin CD207)
Connective tissue	Histiocytes
Kidney	Mesangial cells
Placenta	Hofbauer cells

Granulomatous Inflammation

- Granuloma : focus of chronic inflammation consisting of aggregates of activated macrophages (epithelioid cells) ± giant cells, rim of lymphocytes ± plasma cells, ± central necrosis.
- Epithelioid cells : activated macrophages with abundant pink cytoplasm, slipper/shoe-sole shaped nuclei.
- Key mediators : IFN-γ (Th1) converts macrophages to epithelioid cells; TNF-α maintains granuloma (anti-TNF therapy → TB reactivation).

Types of giant cells :

Giant cell	Features	Seen in
Langhans giant cell	Nuclei in horseshoe / peripheral arrangement	TB, sarcoidosis
Foreign body giant cell	Nuclei scattered haphazardly	Sutures, talc, foreign bodies
Touton giant cell	Ring of nuclei + foamy (lipid) cytoplasm outside ring	Xanthoma, xanthogranuloma, fat necrosis
Reed-Sternberg cell	Owl-eye nucleoli, bilobed nucleus	Hodgkin lymphoma
Aschoff giant cell	Multinucleate Anitschkow cells	Rheumatic heart disease
Warthin-Finkeldey cell	Grape-like nuclei	Measles, lymphoid tissue
Tumour giant cells	Pleomorphic, hyperchromatic	Malignant tumours
Osteoclast-like giant cells	Many nuclei, uniform	Giant cell tumour of bone

Types of granuloma & examples :

Category	Examples
Caseating	Tuberculosis, histoplasmosis, coccidioidomycosis
Non-caseating	Sarcoidosis, Crohn disease, berylliosis, leprosy (tuberculoid), foreign body
Suppurative (stellate)	Cat-scratch disease (Bartonella henselae), LGV (Chlamydia L1-L3), tularemia, yersinia, melioidosis
Foreign body	Sutures, talc (IV drug use), silica
Immune granuloma	TB, leprosy, syphilis, sarcoid, schistosomiasis
Necrotizing granulomatous vasculitis	Granulomatosis with polyangiitis (Wegener)

Category	Examples
Palisading granuloma	Rheumatoid nodule, granuloma annulare, necrobiosis lipoidica

Mnemonic : Causes of granulomas — "TB SCHLEP"

Tuberculosis · Berylliosis, Brucella · Sarcoidosis, Syphilis, Schistosoma · Crohn, Cat-scratch · Histoplasma & other fungi · Leprosy, LGV · EGPA & GPA (vasculitis) · Parasites, foreign Particles (talc, sutures, silica).

Filled : Rank boosters — granuloma

- Granulomatous mastitis, orchitis — idiopathic types.
- Hepatic granulomas : TB, sarcoidosis, drugs (allopurinol, quinidine), Q fever (fibrin-ring / doughnut granuloma), PBC.
- Fibrin-ring (doughnut) granuloma : Q fever (Coxiella), also Hodgkin, CMV, EBV, allopurinol.
- Chronic granulomatous disease — granulomas due to failure of killing, not a "granuloma disease" in the classical sense.
- Sarcoid granulomas : asteroid bodies, Schaumann bodies, Hamazaki-Wesenberg bodies; ACE ↑, hypercalcemia (1α-hydroxylase in macrophages).

Systemic Effects of Inflammation (Acute Phase Response)

1. Fever — pyrogens (LPS → IL-1, TNF → PGE_2 in hypothalamus)
2. Acute phase proteins (liver, IL-6) :
   - C-reactive protein (CRP) — opsonin, activates complement (pentraxin)
   - Fibrinogen → ↑ ESR (rouleaux formation)
   - Serum amyloid A (SAA) — AA amyloid
   - Hepcidin — ↓ iron absorption/release → anemia of chronic disease
   - Ferritin, haptoglobin, ceruloplasmin, α1-antitrypsin, complement
   - Negative acute phase reactants : albumin, transferrin, transthyretin
3. Leukocytosis (↑ release from marrow — IL-1, TNF; later CSFs) :
   - Neutrophilia — bacterial infection
   - Lymphocytosis — viral (mumps, rubella), pertussis, TB
   - Eosinophilia — asthma, allergy, parasites
   - Leukopenia — typhoid, some viruses, rickettsiae, protozoa
   - Leukemoid reaction : > 40,000-100,000/μL
4. ↑ heart rate, BP; ↓ sweating; rigors; anorexia; somnolence
5. Septic shock — TNF, IL-1 → DIC, hypotension, metabolic disturbances

Exam pearl

- ESR is ↑ by fibrinogen (most), globulins; ↓ by sickle cells, polycythemia, spherocytes, hypofibrinogenemia, CHF.
- Procalcitonin ↑ specifically in bacterial sepsis (not viral).
- hs-CRP : cardiovascular risk marker.

TISSUE REPAIR & HEALING

Overview

- Repair = restoration of tissue architecture and function after injury.
- Two processes : Regeneration (replacement by same cells) and Scar formation (fibrosis) (connective tissue deposition).

Cell cycle & proliferative capacity :

Tissue type	Cell cycle	Examples
Labile (continuously dividing)	Continuously cycling	Surface epithelia (skin, GIT, urinary), bone marrow / hematopoietic cells, endometrium
Stable (quiescent)	G0, enter G1 when stimulated	Liver, kidney, pancreas, endothelium, fibroblasts, smooth muscle
Permanent	Terminally differentiated, non-dividing	Neurons, cardiac muscle, (skeletal muscle – limited via satellite cells)

Cell cycle checkpoints

G1 – restriction point (Rb) : most important, cyclin D-CDK4/6 phosphorylates Rb → releases E2F

↓

G1/S checkpoint – p53 → p21 halts damaged cells (DNA repair or apoptosis)

↓

S (DNA synthesis) → G2/M checkpoint (DNA damage)

↓

M (mitosis) – spindle assembly checkpoint

Stem cells :

- Embryonic stem cells – pluripotent; induced pluripotent stem cells (iPSC) created by Oct3/4, Sox2, KLF4, c-MYC (Yamanaka factors).
- Adult (tissue) stem cell niches :

Tissue	Niche
Skin	Bulge of hair follicle, basal layer
Intestine	Crypt base (above Paneth cells; Lgr5+)
Liver	Canals of Hering (oval cells)
Cornea	Limbus
Brain	Subventricular zone, dentate gyrus
Muscle	Satellite cells
Bone marrow	Hematopoietic (CD34+) and mesenchymal stem cells

Growth factors :

Factor	Main actions
EGF / TGF-α	Epithelial and fibroblast mitogen; re-epithelialization, granulation tissue
HGF (scatter factor)	Hepatocyte, epithelial proliferation; cell motility (c-MET receptor)

Factor	Main actions
VEGF	Angiogenesis (most important), ↑ permeability; induced by hypoxia (HIF-1α)
PDGF	Chemotactic for neutrophils, macrophages, fibroblasts, SMC; fibroblast proliferation
FGF (basic, FGF-2)	Angiogenesis, wound contraction, ECM synthesis, migration
TGF-β	Most important fibrogenic agent; ↑ collagen, fibronectin; ↓ MMP; anti-inflammatory
KGF (FGF-7)	Keratinocyte migration, proliferation
Angiopoietin 1/2	Vessel maturation and stabilization (Tie2)

Extracellular Matrix (ECM)

Component	Key facts
Collagen	Most abundant protein. Type I — skin, bone, tendon (most common); type II — cartilage; type III — granulation tissue, blood vessels (reticulin); type IV — basement membrane; type VII — anchoring fibrils
Elastin	Fibrillin-1 scaffold (Marfan)
Fibronectin	Adhesion of cells to ECM; granulation tissue, provisional matrix
Laminin	Most abundant glycoprotein in basement membrane
Proteoglycans	Hyaluronan (cushioning), heparan sulfate
Integrins	Cell-ECM attachment, signalling

Mnemonic : Collagen types — "SCAB"

Type I : Skin, bone, tendon (bone) · Type II : Cartilage (cartwolage) · Type III : Arteries, granulation ("three", reticulin) · Type IV : Basement membrane ("under the floor").

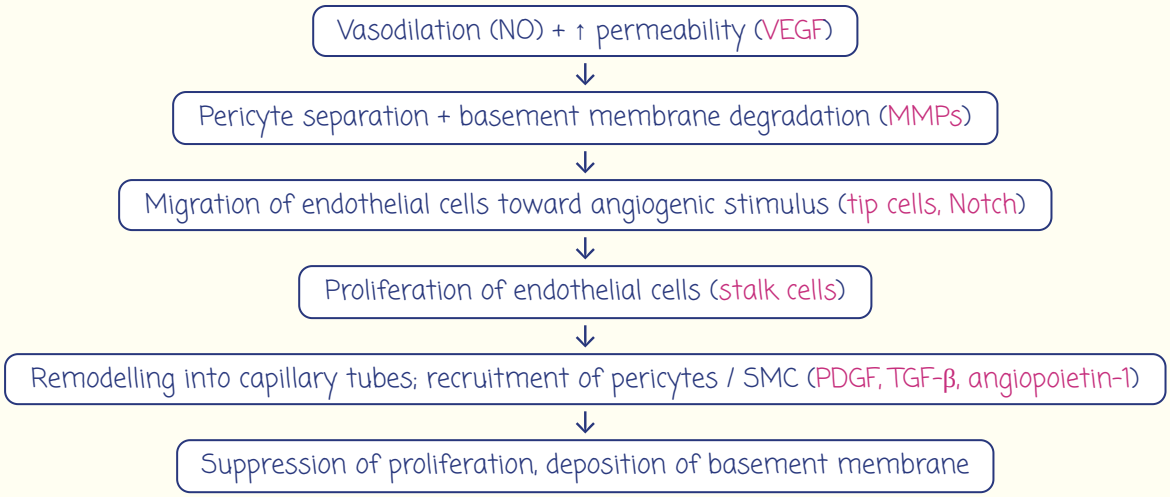
Filled : Collagen synthesis — disorders

Step	Location	Defect
Hydroxylation of proline/lysine	RER	Scurvy (vitamin C)
Glycosylation, triple helix (procollagen)	RER	Osteogenesis imperfecta (COL1A1/A2)
Cleavage of terminal propeptides	Outside cell	Ehlers-Danlos (dermatosparaxis type)
Cross-linking by lysyl oxidase (Cu^{2+})	Outside cell	Menkes disease (ATP7A), EDS
- Alport syndrome — type IV collagen (α5, X-linked). Vascular EDS — type III collagen (COL3A1) — arterial / uterine rupture.		

Liver Regeneration

- After partial hepatectomy (up to 60%) liver regenerates by compensatory hyperplasia of remaining hepatocytes (priming : TNF, IL-6 → G0→G1; growth factors : HGF, TGF-α → progression).
- If hepatocyte proliferation impaired (chronic injury) → regeneration from progenitor (oval) cells in canals of Hering.

Angiogenesis



Scar Formation

Granulation tissue :

- Hallmark of healing: pink, soft, granular gross appearance.
- Microscopy : proliferating fibroblasts + new thin-walled capillaries (angiogenesis) in loose ECM, with macrophages; type III collagen.
- Appears by 3-5 days.

Steps :

1. Angiogenesis
2. Formation of granulation tissue (fibroblast migration and proliferation)
3. Remodelling of connective tissue (type III → type I collagen by MMPs (zinc-dependent); inhibited by TIMPs)

> Exam pearl
> - Myofibroblasts (α-SMA +) → wound contraction (most important in secondary intention).
> - Tensile strength : 10% at 1 week; 70-80% at 3 months (maximum, never reaches 100%).
> - Fibronectin forms provisional scaffold; collagen type III first, then type I.

Cutaneous Wound Healing

Feature	Primary union (first intention)	Secondary union (second intention)
Wound	Clean, surgical, edges approximated	Large, gaping, tissue loss, infection
Inflammation	Less	More intense
Granulation tissue	Scanty	Abundant
Wound contraction	Absent / minimal	Prominent (myofibroblasts)
Scar	Thin, linear	Large, may be deforming
Infection risk	Low	High
Healing time	Short	Long

Timeline of healing by first intention :

Time	Event
24 h	Neutrophils at margins; blood clot (fibrin) + scab; basal cells begin mitosis

Time	Event
24–48 h	Epithelial cells migrate from edges, fuse in midline under scab → thin continuous layer
Day 3	Neutrophils replaced by macrophages; granulation tissue invades incision; collagen fibres vertical, do not bridge
Day 5	Neovascularization maximal; collagen fibrils bridge the incision; epidermis normal thickness
Week 2	Continued collagen accumulation, fibroblast proliferation; "blanching" (↓ vessels, ↓ edema)
End of 1 month	Cellular connective tissue without inflammatory cells, covered by intact epidermis; dermal appendages lost permanently

Factors Affecting Healing

Local :

1. Infection — most important cause of delayed healing
2. Mechanical factors — early motion, tension
3. Foreign bodies — sutures, fragments
4. Size, location and type of wound (face heals faster; poor blood supply delays)
5. Ischemia / poor perfusion (atherosclerosis, venous stasis)

Systemic :

1. Nutrition — protein deficiency, vitamin C deficiency (scurvy), zinc deficiency
2. Glucocorticoids — ↓ TGF-β, ↓ collagen, anti-inflammatory (but useful to ↓ fibrosis in corneal infections)
3. Diabetes mellitus — microangiopathy, ↓ neutrophil function, infection
4. Age, anemia, uremia, malignancy, chemo- and radiotherapy

Complications of wound healing :

Complication	Key features
Deficient scar	Wound dehiscence (abdominal surgery — vomiting, coughing, ileus), ulceration
Hypertrophic scar	Excess collagen, confined to wound margins, regresses over time
Keloid	Collagen extends beyond original wound margins, does not regress, recurs after excision; thick, glassy, hyalinized type I and III collagen bundles; African descent, earlobes, sternum
Exuberant granulation (proud flesh)	Protrudes above skin, blocks re-epithelialization — cauterize / excise
Desmoid (aggressive fibromatosis)	Locally aggressive fibroblast proliferation; β-catenin / APC (Gardner syndrome)
Contracture	Exaggerated contraction — burns, palms, soles, anterior thorax
Marjolin ulcer	Squamous cell carcinoma in chronic scar/burn

Filled

- Keloid — ↑ TGF-β; treatment : intralesional triamcinolone, silicone sheeting, radiotherapy after excision.
- Mast cells are numerous in keloids (pruritus).
- Fibrosis in parenchymal organs (cirrhosis, IPF) driven by TGF-β — "persistent repair".
- Hepatic stellate (Ito) cells → myofibroblasts → collagen in space of Disse (cirrhosis).

Healing at Specific Sites

Fracture healing :

Hematoma formation (fibrin clot) — immediate, platelets release PDGF, TGF-β, FGF

↓

Soft (procallus) callus — granulation tissue; osteoprogenitor cells; end of 1st week

↓

Hard (bony) callus / provisional callus — woven bone + hyaline cartilage; maximum at end of 2-3 weeks

↓

Remodelling — woven → lamellar bone along lines of weight-bearing (osteoclasts + osteoblasts); months

- Complications : malunion, delayed union, non-union → pseudarthrosis (cleft lined by synovial-like cells), infection (open fracture), fat embolism.
- Factors impairing : movement at fracture site, poor blood supply (scaphoid, femoral neck, talus), infection, interposition of soft tissue, nutrition.

Other sites :

Site	Healing features
Nervous system (CNS)	No regeneration of neurons; gliosis by astrocytes (gemistocytes) — not fibroblasts
Peripheral nerve	Wallerian degeneration distal; Schwann cell guidance; regrowth 1-3 mm/day; traumatic (amputation) neuroma if misdirected
Cardiac muscle	Fibrous scar (no regeneration)
Skeletal muscle	Limited regeneration from satellite cells
Liver	Regeneration (if framework intact) or fibrosis
Kidney	Tubular epithelium regenerates (ATN — if basement membrane intact); glomeruli do not
Lung	Type II pneumocytes regenerate alveolar epithelium (stem cells for type I)
Cartilage	Poor healing (avascular) — fibrocartilage
Mucosa (GIT)	Rapid regeneration from crypt stem cells
Cornea	Limbal stem cells

Exam pearl

- Healing of nerve : Bands of Büngner (Schwann cell columns guiding axons).
- Chromatolysis — swelling of neuron cell body, dispersion of Nissl substance, nucleus pushed to periphery — axonal reaction.

HEMODYNAMIC DISORDERS, THROMBOSIS & SHOCK

Edema

- Accumulation of excess fluid in interstitial tissue or body cavities.
- Generalized severe edema = anasarca. In cavities : hydrothorax, hydropericardium, hydroperitoneum (ascites).

Pathophysiologic categories :

Mechanism	Causes
↑ Hydrostatic pressure	Impaired venous return : CHF, constrictive pericarditis, ascites in cirrhosis, venous obstruction (DVT), external pressure; arteriolar dilation (heat)
↓ Plasma oncotic pressure (hypoproteinemia)	Nephrotic syndrome (most common), cirrhosis, protein malnutrition (kwashiorkor), protein-losing gastroenteropathy
Lymphatic obstruction	Inflammatory (filariasis → elephantiasis), neoplastic, post-surgical (axillary dissection → arm edema), post-irradiation; peau d'orange in breast cancer
Sodium retention	Excessive salt intake with renal insufficiency, ↑ renin-angiotensin-aldosterone (CHF), acute GN
Inflammation	Acute & chronic inflammation, angiogenesis (exudate)

Exam pearl

- Renal edema (nephrotic) : generalized, first appears in loose connective tissue — periorbital (eyelids), morning.
- Cardiac edema : dependent (legs in standing, sacrum in recumbent).
- Pulmonary edema : left ventricular failure (most common) — heavy wet lungs (2-3× normal), frothy, blood-tinged fluid on cut section.
- Brain edema : flattened gyri, narrowed sulci, compressed ventricles → herniation.
- Pitting edema — interstitial fluid; non-pitting — lymphedema, myxedema.

Hyperemia & Congestion

Feature	Hyperemia (active)	Congestion (passive)
Mechanism	Arteriolar dilation → ↑ inflow	Impaired venous outflow
Colour	Red (erythema) — oxygenated blood	Blue-red (cyanosis) — deoxygenated
Examples	Exercise, inflammation, blushing	CHF (systemic), venous obstruction (local)

Chronic venous congestion (CVC) — organs :

Organ	Cause	Morphology
Lung	Left heart failure, mitral stenosis	Acute : engorged alveolar capillaries, septal edema. Chronic : thick fibrotic septa, hemosiderin-laden macrophages ("heart failure cells") in alveoli → brown induration of lung
Liver	Right heart failure, (Budd-Chiari)	Nutmeg liver : congested red-brown centrilobular areas (zone 3) alternating with pale/fatty periportal areas. Centrilobular hemorrhagic necrosis; cardiac sclerosis / cardiac cirrhosis (long-standing)
Spleen	Right heart failure, portal hypertension	Congestive splenomegaly (up to 500-1000 g), firm; Gamna-Gandy bodies (siderofibrotic nodules — hemosiderin + calcium on collagen)
Kidney	Right heart failure	Enlarged, cyanotic; medulla congested

Filled

- Heart failure cells — demonstrated with Perls Prussian blue (hemosiderin, blue).
- Zone 3 necrosis : congestion (RHF), shock, acetaminophen, CCl_4, alcohol, halothane. Zone 1 : phosphorus poisoning, eclampsia (periportal fibrin). Zone 2 : yellow fever.

Hemorrhage

Term	Size / site
Petechiae	1-2 mm — ↑ capillary pressure, thrombocytopenia, platelet dysfunction
Purpura	≥ 3 mm (up to 1 cm) — trauma, vasculitis, vascular fragility
Ecchymoses (bruise)	> 1-2 cm subcutaneous hematomas; colour changes : red-blue (Hb) → blue-green (bilirubin) → golden-brown (hemosiderin)
Hematoma	Blood accumulation within tissue
Hemothorax, hemopericardium, hemoperitoneum, hemarthrosis	Blood in cavities

- Clinical significance : volume (> 20% rapid loss → hemorrhagic shock), site (intracranial), chronic → iron deficiency anemia.

Hemostasis — Normal

Arteriolar vasoconstriction (reflex + endothelin)

↓

Primary hemostasis : platelet adhesion (vWF-GpIb) → shape change, granule release (ADP, TXA_2) → aggregation (GpIIb/IIIa - fibrinogen) → platelet plug

↓

Secondary hemostasis : tissue factor (factor III) → coagulation cascade → thrombin → fibrin

↓

Clot stabilization & resorption : fibrin polymerization (factor XIIIa), platelet contraction; t-PA → plasmin limits clot

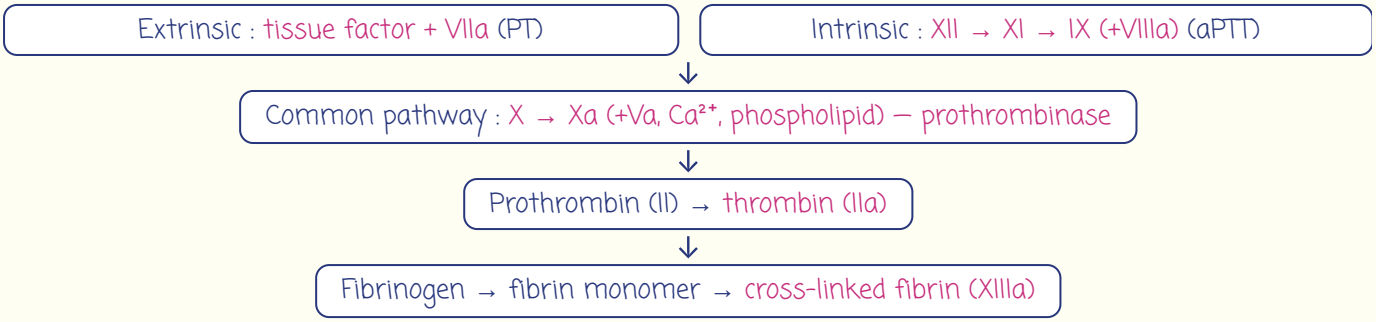
Endothelium — anti- vs pro-thrombotic :

Antithrombotic (normal)	Prothrombotic (injured / activated)
PGI_2, NO (inhibit platelets); ADPase (CD39)	vWF (platelet adhesion)
Thrombomodulin → thrombin activates protein C	Tissue factor (induced by TNF, IL-1, LPS)
Heparin-like molecules → ↑ antithrombin III	PAI-1 (↓ fibrinolysis)
TFPI (tissue factor pathway inhibitor)	
t-PA (fibrinolysis)	

Platelets :

- α-granules : fibrinogen, fibronectin, factor V, vWF, PF4, PDGF, TGF-β, P-selectin.
- Dense (δ) granules : ADP, ATP, Ca^{2+}, serotonin, histamine, epinephrine.
- Disorders : Bernard-Soulier (GpIb) — giant platelets, no ristocetin aggregation; Glanzmann thrombasthenia (GpIIb/IIIa) — no aggregation with ADP, normal ristocetin.

Coagulation cascade :



- Vitamin K-dependent factors : II, VII, IX, X, protein C, protein S (γ-carboxylation of glutamate). Shortest half-life : factor VII (~6 h) → PT prolongs first with warfarin.
- Thrombin : most important — converts fibrinogen, activates V, VIII, XI, XIII, platelets (PAR receptors); with thrombomodulin activates protein C.
- Natural anticoagulants : antithrombin III (inhibits IIa, IXa, Xa, XIa, XIIa; ↑ by heparin), protein C + S (inactivate Va and VIIIa), TFPI.

Thrombosis

Virchow's triad :

1. Endothelial injury — most important; dominant in heart and arterial circulation (MI, ulcerated plaque, vasculitis, hypertension, toxins of smoking)
2. Abnormal blood flow (stasis or turbulence) — stasis : venous thrombi; turbulence : arterial and cardiac thrombi
3. Hypercoagulability (thrombophilia)

Hypercoagulable states :

Primary (genetic) :

1. Factor V Leiden (Arg506Gln; resistance to activated protein C) — most common inherited cause
2. Prothrombin G20210A mutation (↑ prothrombin) — second most common
3. Deficiency of antithrombin III (heparin resistance), protein C, protein S
4. ↑ factor VIII, IX, XI, fibrinogen
5. Hyperhomocysteinemia (MTHFR — also acquired : folate / B12 / B6 deficiency)

Secondary (acquired) :

1. Prolonged bed rest / immobilization, surgery, fractures, burns
2. Malignancy (Trousseau syndrome — migratory thrombophlebitis; mucin-secreting adenocarcinomas — pancreas)
3. Antiphospholipid antibody syndrome
4. Heparin-induced thrombocytopenia (HIT) — antibodies to heparin-PF4 complex, day 5–14
5. Pregnancy & postpartum, oral contraceptives, estrogen therapy
6. Nephrotic syndrome (loss of antithrombin III)
7. Myeloproliferative neoplasms (polycythemia vera, JAK2), PNH, sickle cell
8. Smoking, obesity, cardiomyopathy, atrial fibrillation, prosthetic valves, DIC

Filled : Antiphospholipid syndrome

- Antibodies : lupus anticoagulant, anticardiolipin, anti-β2-glycoprotein I.
- In vitro : prolonged aPTT (paradoxically) ; in vivo : thrombosis.
- Features : recurrent arterial/venous thrombosis, recurrent miscarriages, thrombocytopenia, false-positive VDRL, livedo reticularis.
- Catastrophic APS — multiorgan microvascular thrombosis.
- Treatment : lifelong warfarin (not DOACs in triple-positive).

Morphology of thrombi :

- Lines of Zahn — alternating pale layers of platelets + fibrin and dark layers of RBCs — indicate thrombosis in flowing blood (antemortem).

Feature	Antemortem thrombus	Postmortem clot
Lines of Zahn	Present	Absent
Attachment	Firmly attached to vessel wall	Not attached; gelatinous
Appearance	Dry, granular, friable, grey-red	Rubbery, dark red "currant jelly" dependent portion + yellow "chicken fat" supernatant
Shape	Shape of vessel ± tail	Cast of vessel

Type	Features
Arterial / cardiac (mural)	Endothelial injury/turbulence; begin at injury; grow retrograde; white / pale (platelet-rich); occlusive in coronary, cerebral, femoral arteries
Venous (phlebothrombosis)	Stasis; grow in direction of flow (toward heart); red / stasis thrombi (RBC-rich); deep veins of leg (90%)
Vegetations	Thrombi on heart valves — infective endocarditis, NBTE (marantic), Libman-Sacks (SLE)

Fate of thrombus :

1. Propagation — accumulates more platelets, fibrin → obstruction
2. Embolization — dislodges, travels
3. Dissolution — fibrinolysis (recent thrombi)
4. Organization and recanalization — ingrowth of endothelial cells, SMC, fibroblasts; new capillary channels
5. Septic infection → mycotic aneurysm

> Exam pearl
> - Most common site of DVT : deep veins of calf; most significant (embolism) : popliteal, femoral, iliac veins (above knee).
> - Superficial thrombophlebitis (saphenous varicosities) rarely embolize.
> - Phlegmasia alba dolens (milk leg) — iliofemoral thrombosis in pregnancy.

Disseminated Intravascular Coagulation (DIC)

- Consumption coagulopathy : widespread microvascular thrombi → consumption of platelets and factors → bleeding.

Causes :

1. Obstetric complications — abruptio placentae, retained dead fetus, amniotic fluid embolism, septic abortion, eclampsia
2. Infections — Gram-negative sepsis (most common), meningococcemia, malaria, Rocky Mountain spotted fever
3. Neoplasms — acute promyelocytic leukemia (M3), adenocarcinomas (pancreas, prostate, lung)
4. Massive tissue injury — trauma, burns, extensive surgery
5. Others — snake bite, acute intravascular hemolysis, shock, giant hemangioma (Kasabach-Merritt), heat stroke

Release of tissue factor (placenta, tumour, APL granules) | Endothelial injury (sepsis, LPS, TNF)

↓

Widespread thrombin generation → microthrombi (fibrin) in small vessels

↓

Consumption of platelets, fibrinogen, V, VIII → bleeding | Microangiopathic hemolytic anemia (schistocytes), ischemic organ damage

↓

Secondary fibrinolysis → ↑ D-dimer, FDPs (most specific : D-dimer)

- Lab : ↓ platelets, ↑ PT and aPTT, ↓ fibrinogen, ↑ D-dimer, schistocytes.
- Organs : kidney (cortical necrosis), adrenal (Waterhouse-Friderichsen in meningococcemia), pituitary (Sheehan), brain, lungs.

Embolism

- Detached intravascular solid, liquid or gaseous mass carried by blood to a distant site. 99% from thrombi (thromboembolism).

Pulmonary thromboembolism :

1. Source : 95% from deep leg veins (above knee)
2. 60-80% clinically silent (small)
3. Occluding > 60% of pulmonary circulation → sudden death, acute right heart failure (cor pulmonale), cardiovascular collapse
4. Saddle embolus — at bifurcation of main pulmonary artery
5. Small-medium artery obstruction → hemorrhage (bronchial circulation intact); infarction only when circulation compromised (LVF)
6. Paradoxical embolism — venous embolus via ASD / PFO into systemic circulation
7. Multiple emboli over time → pulmonary hypertension

Systemic thromboembolism :

- 80% from intracardiac mural thrombi (2/3 due to LV wall infarcts, 1/4 left atrial dilation + atrial fibrillation).
- Sites : lower extremities (75%), brain (10%), intestines, kidneys, spleen.

Other emboli :

Type	Key features
Fat & marrow embolism	Long bone fractures (tibia, femur), burns, CPR; syndrome after 1-3 days : pulmonary insufficiency, neurologic symptoms, anemia, thrombocytopenia, petechial rash (upper body, axilla); Oil Red O / Sudan stain on frozen section; fatal in ~10%
Air embolism	> 100 cc needed for clinical effect; obstetric procedures, chest wall injury, IV lines. Decompression sickness (caisson disease) — nitrogen bubbles : bends (joints), chokes (lungs); chronic : aseptic necrosis of femoral head, tibia, humerus
Amniotic fluid embolism	Mortality ~80%; sudden dyspnea, cyanosis, shock, seizures, coma, DIC; fetal squamous cells, lanugo hair, fat (vernix), mucin in maternal pulmonary microcirculation
Tumour, bone marrow, atheromatous (cholesterol), foreign body (talc), septic	Atheroembolism : cholesterol clefts, after angiography; livedo reticularis, renal failure, eosinophilia

Infarction

- Area of ischemic necrosis caused by occlusion of either arterial supply or venous drainage. 99% due to thrombotic or embolic events.

Feature	Red (hemorrhagic) infarct	White (anemic / pale) infarct
Circulation	Venous occlusion; dual blood supply; loose tissues; previously congested tissue; reperfusion	Arterial occlusion in solid organs with end-arterial circulation
Examples	Lung, small intestine, ovarian torsion, testicular torsion, liver (dual), brain (embolic, reperfusion)	Heart, spleen, kidney
Shape	Wedge-shaped with occluded vessel at apex, periphery as base	
Histology	Coagulative necrosis (except brain : liquefactive)	Coagulative necrosis

Factors affecting development of infarct :

1. Anatomy of vascular supply — dual (lung, liver) or parallel (radial + ulnar artery) supply protects; end-arteries (kidney, spleen) do not
2. Rate of occlusion — slow occlusion allows collaterals (coronary)
3. Tissue vulnerability to hypoxia — neurons (3-4 min) > myocardium (20-30 min) > fibroblasts
4. Oxygen content of blood — anemia, cyanosis, CHF worsen

- Septic infarction → abscess (infected emboli from IE).
- Earliest histologic change in infarct : none in 4-12 h; acute inflammation at margins within hours to 1-2 days.

Gangrene

Feature	Dry gangrene	Wet gangrene	Gas gangrene
Mechanism	Arterial occlusion, slow	Venous + arterial occlusion, infection (putrefaction)	Clostridium perfringens (α-toxin, lecithinase)
Site	Limbs (toes)	Bowel (strangulated hernia, volvulus), diabetic foot, lung	Contaminated deep wounds, muscle (myonecrosis)
Line of demarcation	Clear	Poorly defined	Absent
Appearance	Dry, shrunken, black (iron sulfide from Hb)	Moist, swollen, foul-smelling	Crepitus, gas bubbles
Spread	Slow	Rapid	Very rapid
Prognosis	Good	Poor (septicemia)	Poor

- Examples : Fournier gangrene (scrotum/perineum — synergistic), noma (cancrum oris) (malnourished children), Meleney's synergistic gangrene.

Shock

- State of cellular hypoperfusion due to ↓ cardiac output or ↓ effective circulating blood volume → impaired tissue perfusion and hypoxia.

Type	Mechanism	Examples
Cardiogenic	Myocardial pump failure	MI, arrhythmias, cardiac tamponade, pulmonary embolism (obstructive)
Hypovolemic	Inadequate blood or plasma volume	Hemorrhage, burns, vomiting, diarrhea
Septic (distributive)	Peripheral vasodilation + pooling, endothelial activation/injury, DIC	Gram-negative > Gram-positive sepsis, fungi

Type	Mechanism	Examples
Neurogenic	Loss of vascular tone	Spinal cord injury, anesthetic accident
Anaphylactic	IgE-mediated systemic vasodilation, ↑ permeability	Drugs, insect stings, foods

Pathogenesis of septic shock :

1. Inflammatory mediators — PAMPs (LPS) → TLRs → TNF, IL-1, IL-6, IFN-γ, HMGB1; complement C3a, C5a
2. Endothelial activation and injury — capillary leak, edema, ↑ tissue factor → DIC
3. Immune suppression — CARS (compensatory anti-inflammatory response) — IL-10, lymphocyte apoptosis
4. Metabolic abnormalities — insulin resistance, hyperglycemia; relative adrenal insufficiency
5. Organ dysfunction — ARDS, AKI, liver, heart (↓ contractility)

Stages of shock :

Nonprogressive (compensated) : ↑ sympathetic (tachycardia, peripheral vasoconstriction, cool skin), RAAS, ADH — vital organ perfusion maintained

↓

Progressive : tissue hypoperfusion → anaerobic glycolysis → lactic acidosis → arteriolar dilation, blood pooling; DIC, ARDS

↓

Irreversible : lysosomal enzyme leakage, myocardial depression, ischemic bowel → bacteremia; ATN — death

Morphology of shock :

Organ	Change
Brain	Ischemic encephalopathy — watershed infarcts
Heart	Subendocardial hemorrhage, necrosis, contraction band necrosis
Kidney	Acute tubular necrosis (most common cause of oliguria)
Lung	Relatively resistant in hypovolemic shock; DAD (shock lung, ARDS) in septic shock — hyaline membranes
Adrenal	Lipid depletion of cortical cells (stress); Waterhouse-Friderichsen hemorrhage
GIT	Patchy mucosal hemorrhages, necrosis (hemorrhagic enteropathy)
Liver	Fatty change, central (zone 3) hemorrhagic necrosis

Filled : Warm vs cold shock

- Septic shock — early "warm" shock: vasodilation, warm flushed skin, ↑ cardiac output.
- Cardiogenic / hypovolemic — "cold" shock : cool clammy skin, ↓ CO, ↑ SVR.
- qSOFA : RR ≥ 22, altered mentation, SBP ≤ 100. Septic shock = vasopressors to keep MAP ≥ 65 + lactate > 2 mmol/L despite fluids.

Fluid & Electrolyte Derangements

Disorder	Key causes	Key features
Hyponatremia (< 135)	SIADH (small cell lung Ca, CNS, drugs), CHF, cirrhosis, diuretics, Addison	Cerebral edema, seizures; too-rapid correction → osmotic demyelination (central pontine myelinolysis)
Hypernatremia (> 145)	Water loss, diabetes insipidus	Brain shrinkage, hemorrhage

Disorder	Key causes	Key features
Hypokalemia	Diuretics, vomiting, hyperaldosteronism, insulin, β-agonists	U waves, flat T, muscle weakness, ileus
Hyperkalemia	Renal failure, Addison, ACEi, tissue lysis, acidosis	Peaked T waves, wide QRS, arrhythmia
Hypocalcemia	Hypoparathyroidism, CKD, vitamin D deficiency, pancreatitis	Tetany (Chvostek, Trousseau), prolonged QT
Hypercalcemia	Hyperparathyroidism, malignancy	"Stones, bones, groans, moans", short QT
Metabolic acidosis (↑ AG)	MUDPILES : methanol, uremia, DKA, paraldehyde, isoniazid/iron, lactic, ethylene glycol, salicylates	Kussmaul breathing
Metabolic alkalosis	Vomiting, diuretics, hyperaldosteronism	Hypokalemia

Mnemonic : Normal anion gap acidosis — "HARDASS"

Hyperalimentation · Acetazolamide · RTA · Diarrhea · Addison · Spironolactone · Saline infusion.